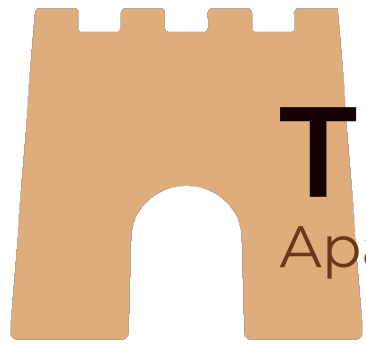




THE MUD CASTLE

Apartment buildings in Karachi Pakistan



CONTEXT

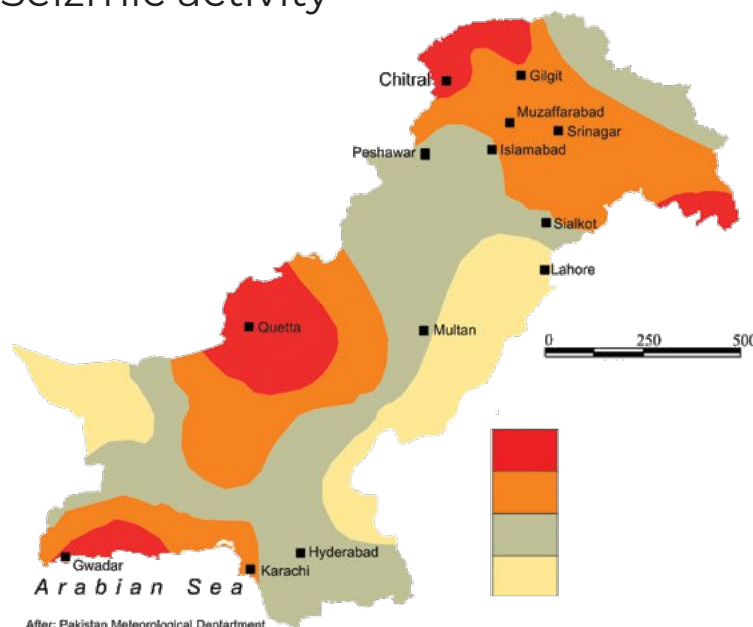
Karachi Pakistan

Extreme conditions:

- Flooding

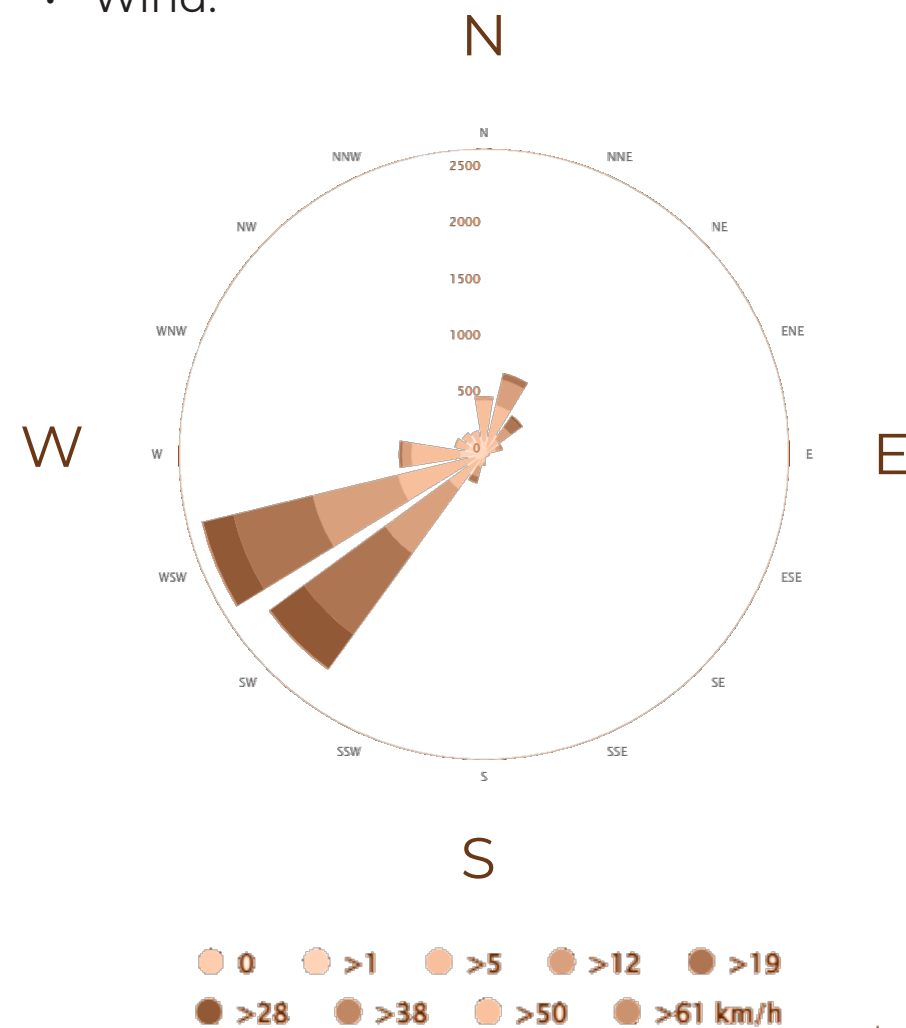


- Seismic activity

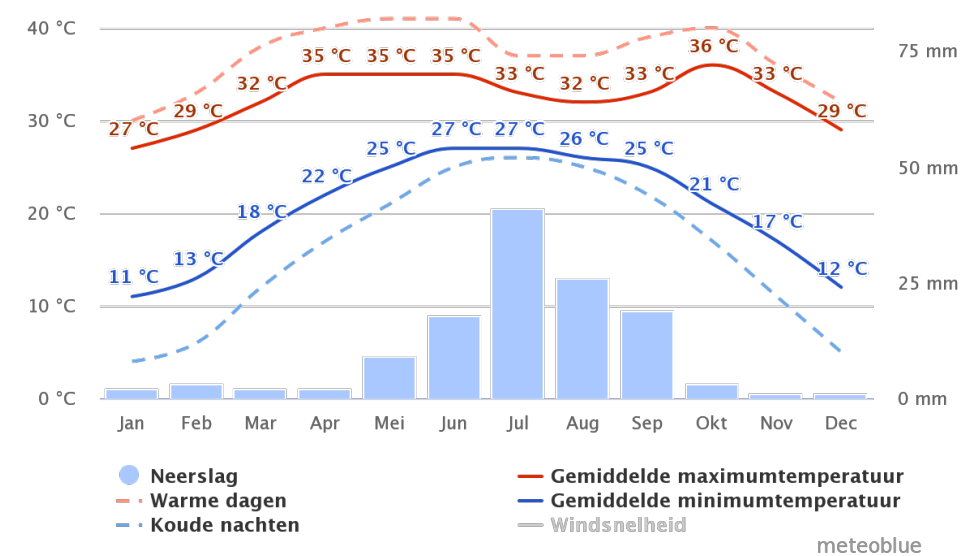


Climate:

- Wind:

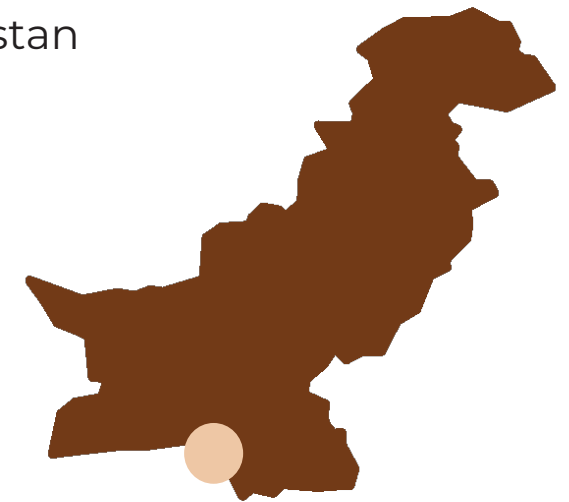


- Annual temperatures and rainfall:



Geography:

- Pakistan



- Province of Sindh



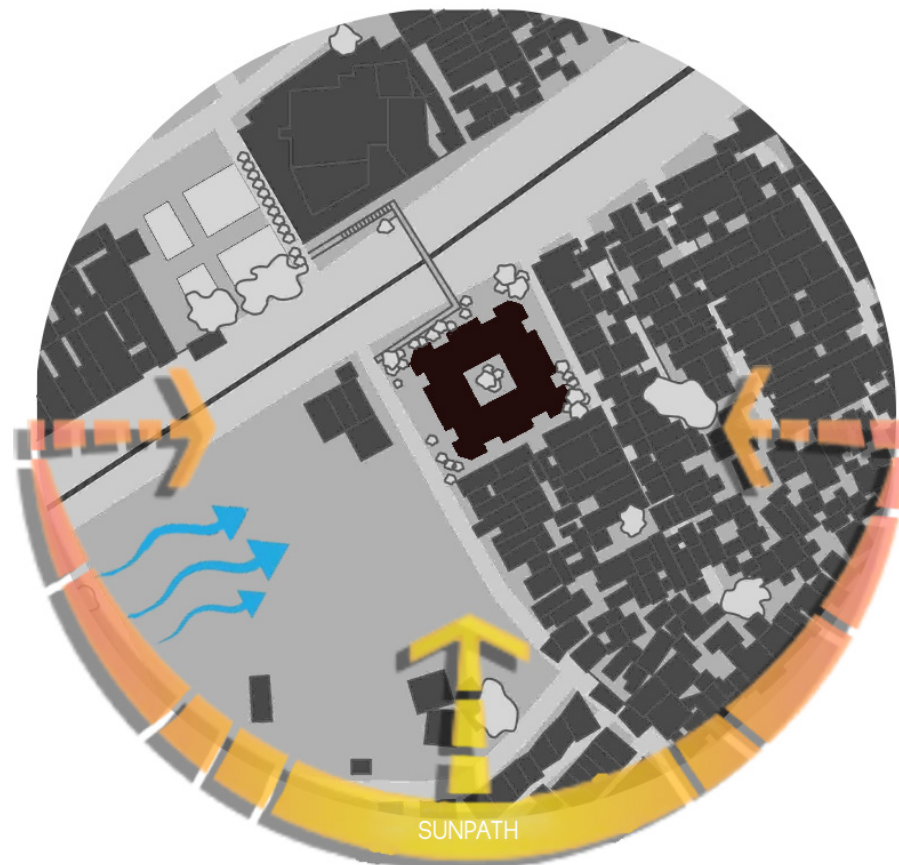
- Karachi



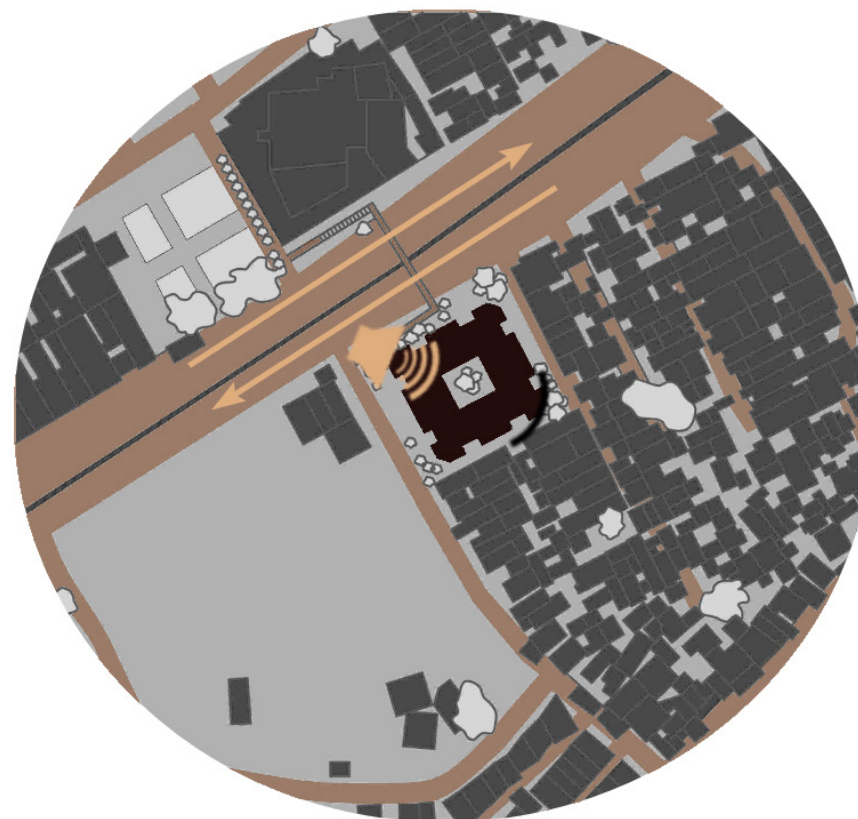


CONTEXT

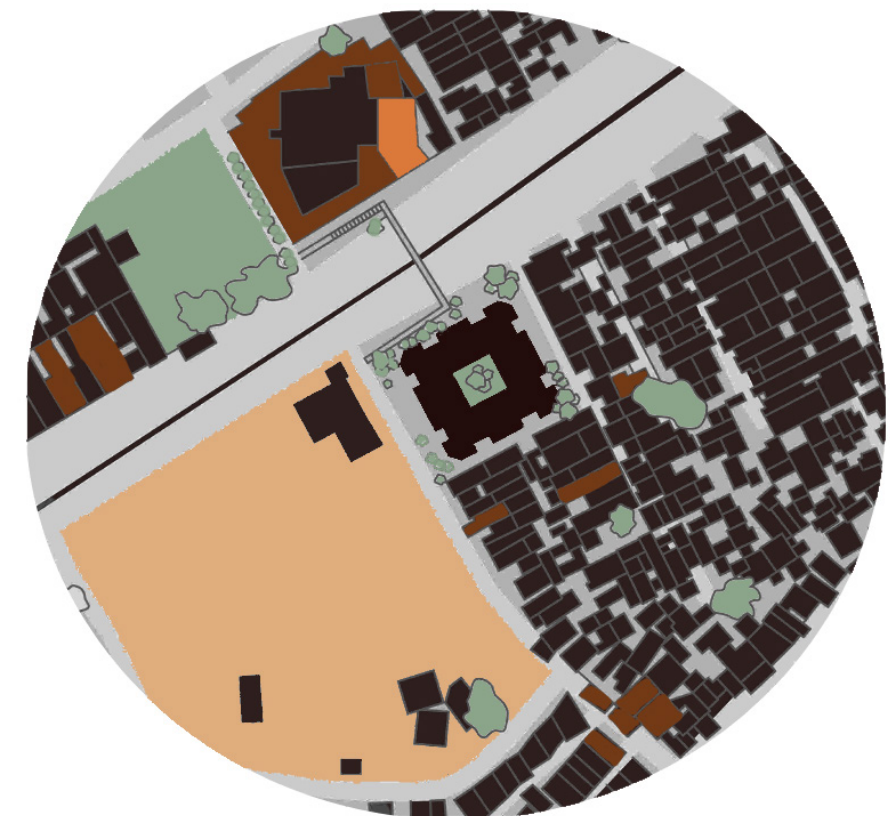
Hafta bazaar New MA Jinnah Road Karachi Sindh Pakistan



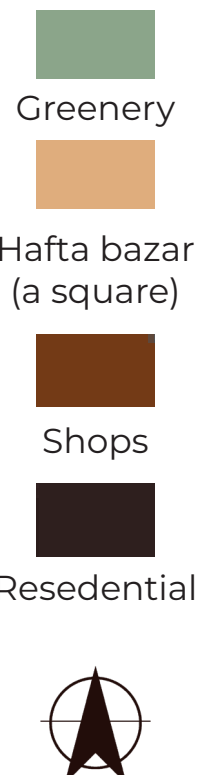
Climate



Main infrastructure

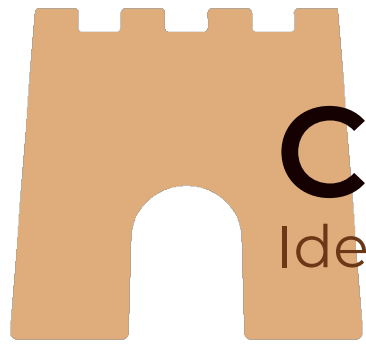


Use of land



Sources photo's:
Tahir Zahid Ishrad



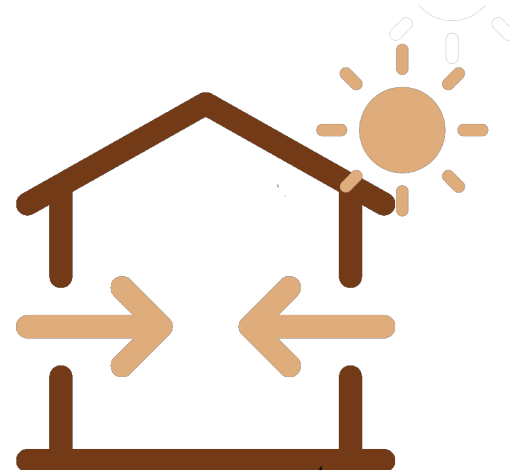


CONCEPT

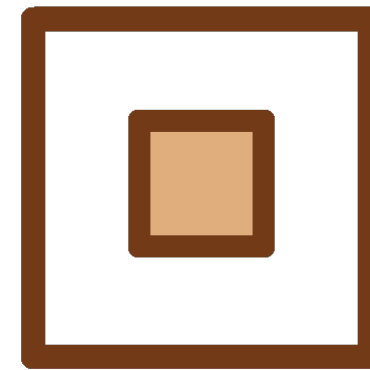
Ideals



Local and sustainable



Passive climatisation



Central courtyard



Faced inwards

CONCEPT

Scetches and idea's

Building like a termite:

- Using adobe (a material made from earth and organic materials) for the facade and load bearing stucture, tapering to the top
- Thermall mass to regulate climate
- And a center courtyard for ventilation

Layout:

- More open at the bottom level, shops and storage at the bottom level, apartments are on the higher levels

Facade:

- Tall and narrow window with rasters (Jharokha or Jali)

Shibam city, Yemen



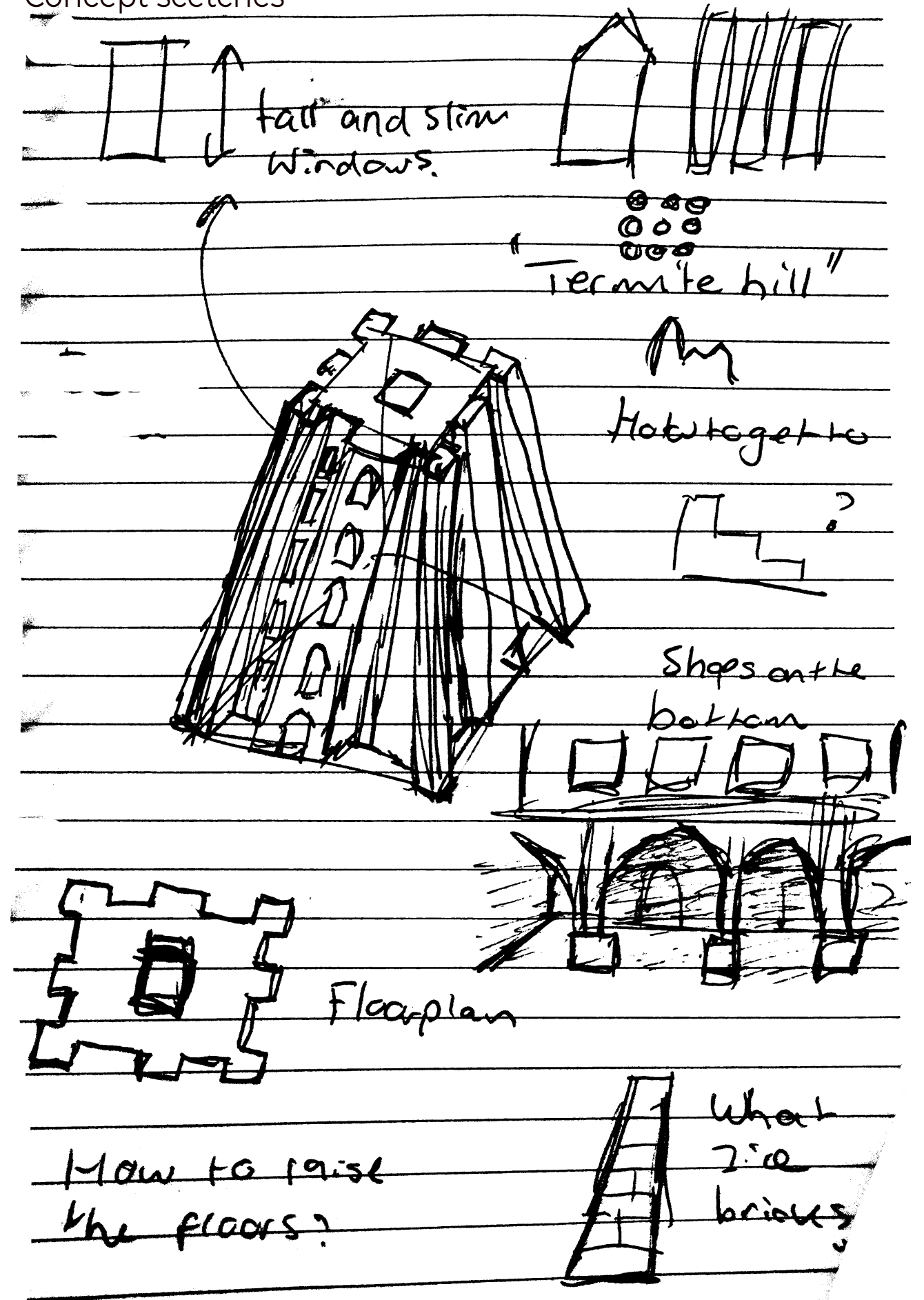
Termite hill



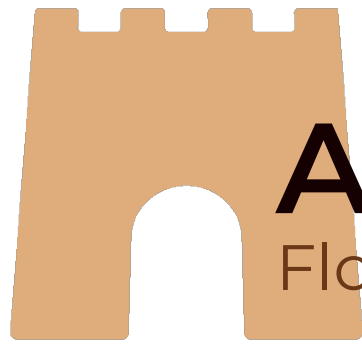
Stone Jali, Amer fort



Concept scetches



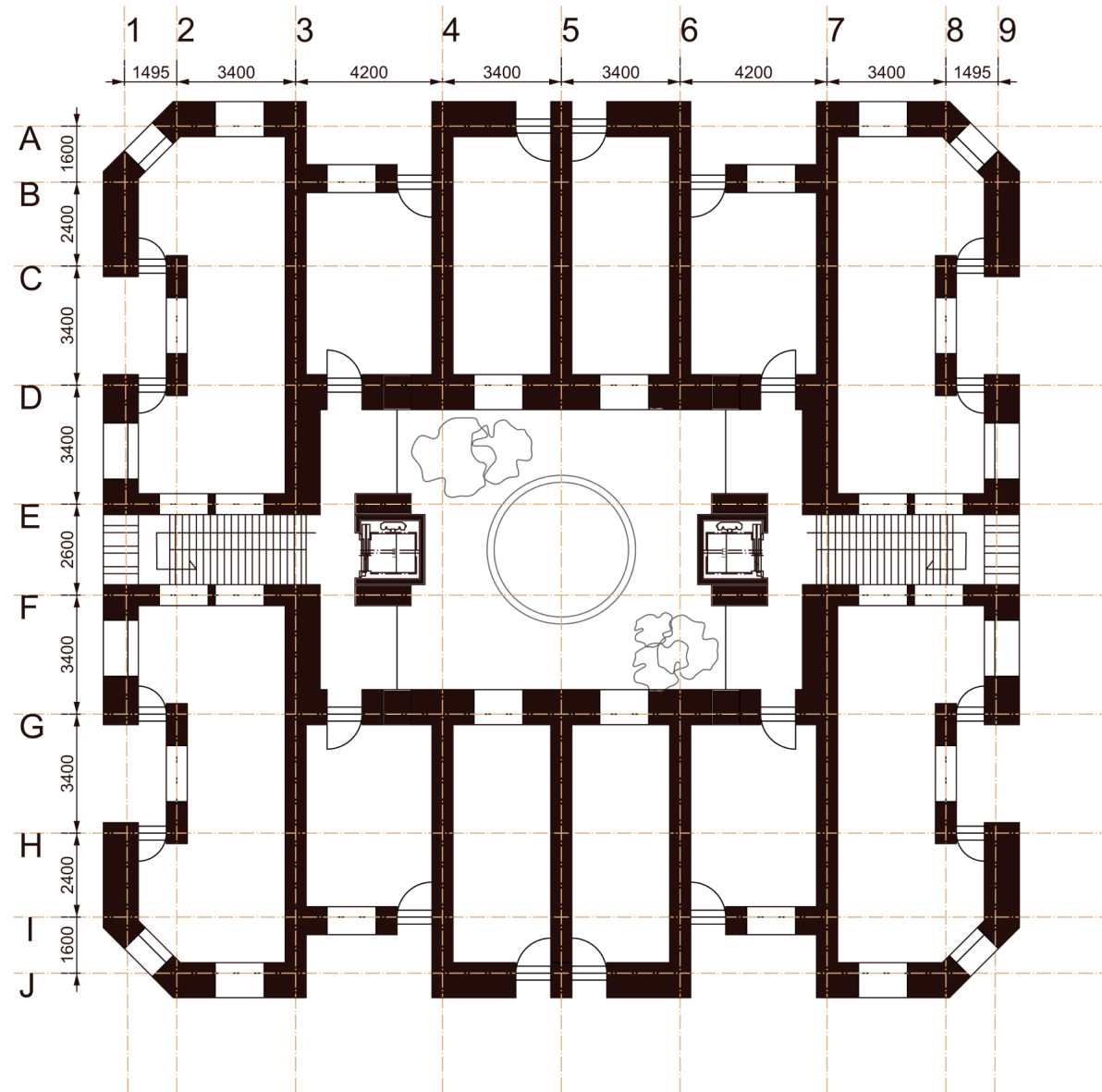
Sources top to bottom:
 Maltagliati, L. (2022). The 2 thousand years old MANHATTAN in the DESERT. Res Humana.
 Fultonk. (2014). The Incredible Termite Mound. Nature.
 Benjamin Parker (2017). Amer Fort, Jaipur



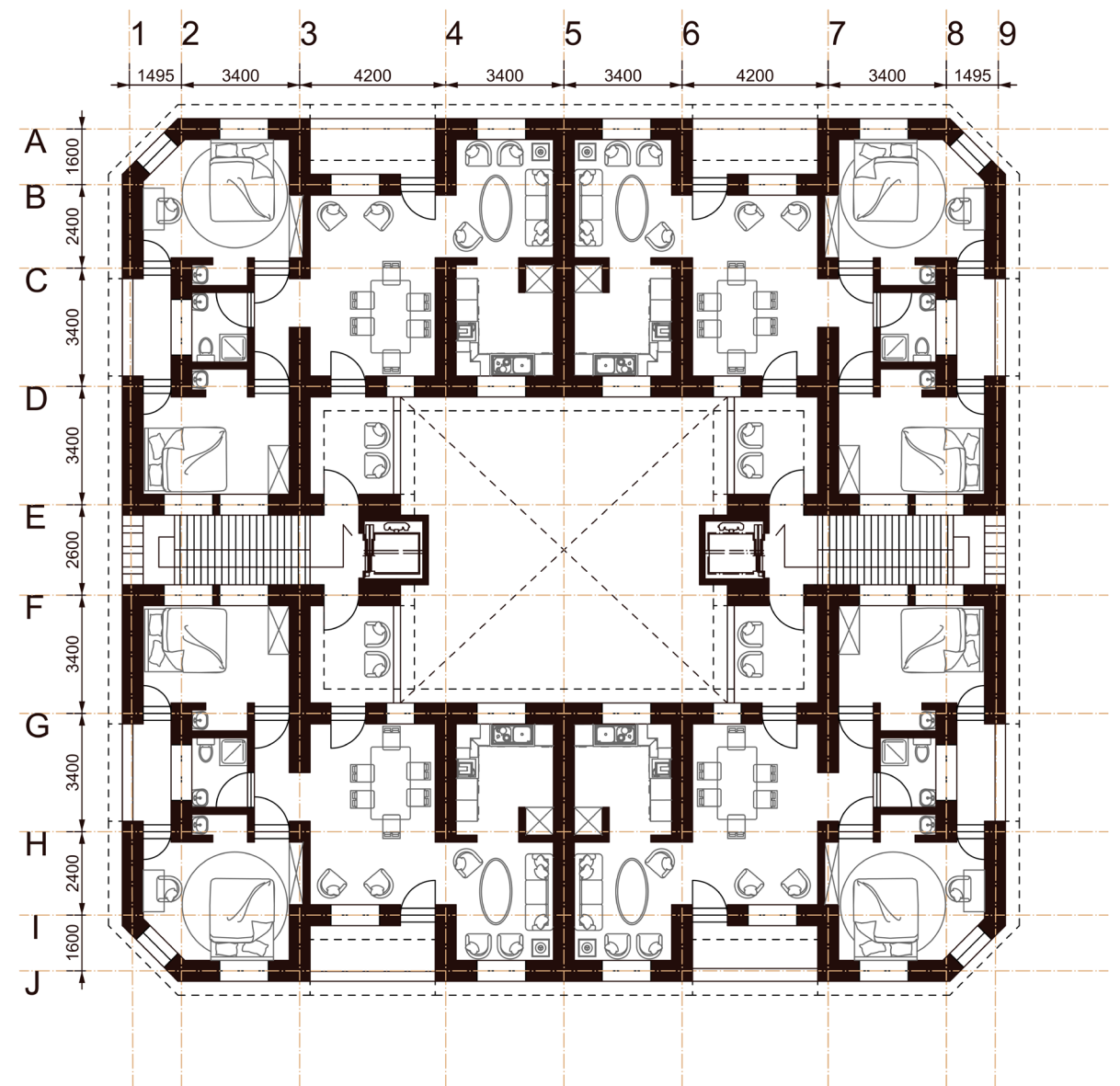
Architectural design

Floorplans

Ground floor



Apartment floors





The floor plan is a square layout divided into a grid of 9 columns (1-9) and 10 rows (A-J). The dimensions for each section are as follows:

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9
A	1495	3400	4200	3400	3400	4200	3400	1495	
B									
C									
D									
E									
F									
G									
H									
I									
J									

The plan includes various icons: shopping carts, boxes, envelopes, and a central circular area with trees. A red path is highlighted, starting from a central point and moving towards the bottom left.

The floor plan is a 10-story building with a grid system. The horizontal axis is labeled 1 through 9, and the vertical axis is labeled A through J. Dimensions are provided for each grid cell:

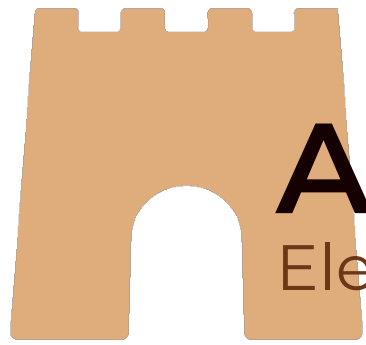
- Horizontal dimensions (top): 1495, 3400, 4200, 3400, 3400, 4200, 3400, 1495.
- Vertical dimensions (left): 1600, 2400, 3400, 3400, 2600, 3400, 3400, 2400, 1600.

The plan shows a central atrium area with a large 'X' mark. A red path is highlighted, starting from the left edge, moving right, then up, then right again, ending at a room in the upper left quadrant.

:Zoning

Private

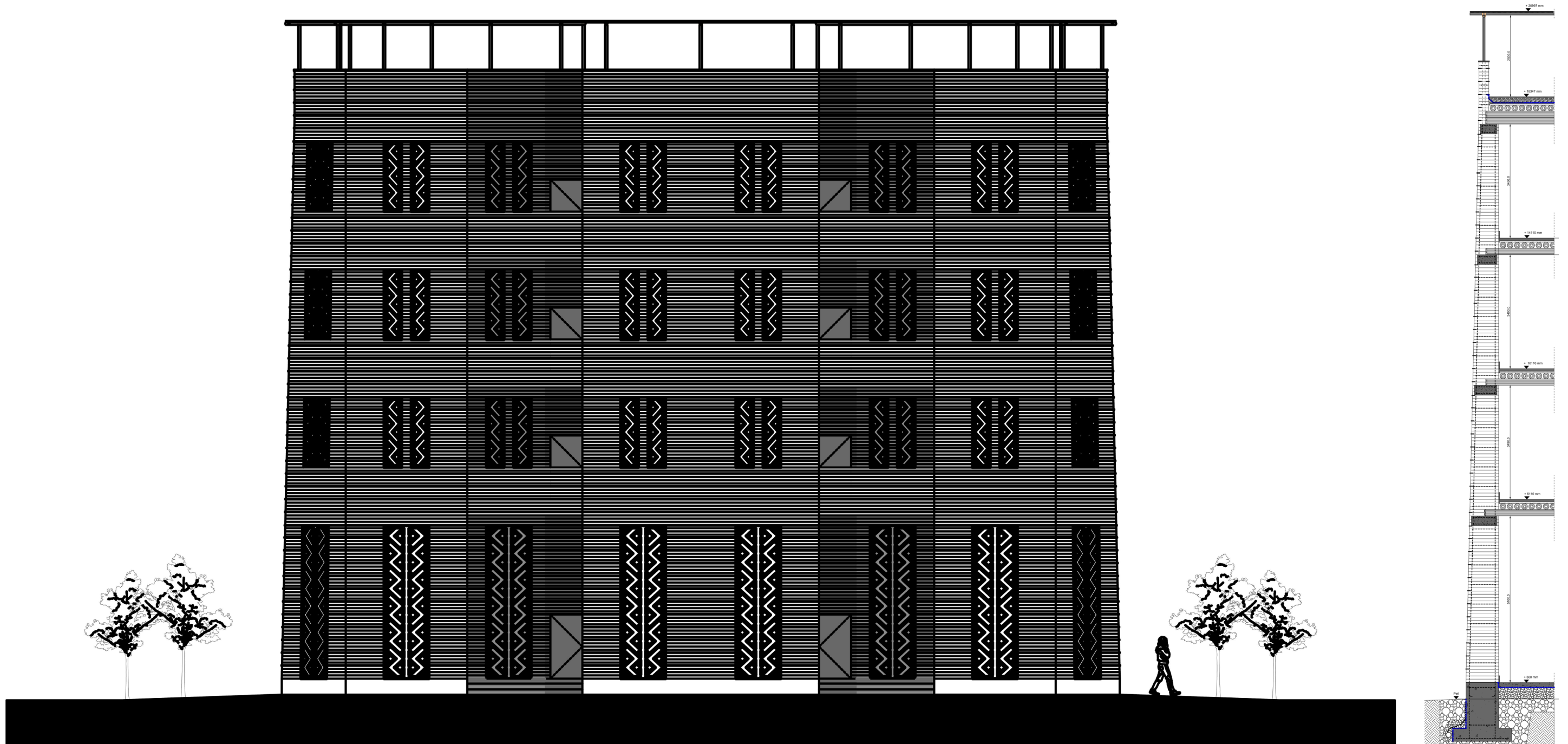




Architectural design

Elevation and section

South/ South East facade





Architectural design

Materiality



Lime stabilized
Steel and bamboo reinforced
Rammed earth
(Using the soil from right
under your feet)



Bamboo floor and roof
construction
Bambusa bamboo
(Cultivated native species)



Concrete ground floor,
Reinforced concrete
ringbeams and foundation



Wooden window frames
and Jali



Fired mudbrick tiles as
floor finish and brick checks

Architectural design

Building sequence



Laying the foundation



Ramming the earth +/-
100mm layers at a time
Adding brick check every 40
mm



Setting the ring beam



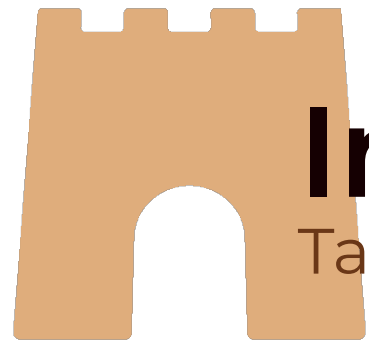
Laying the floor beams



Adding window



Next layer



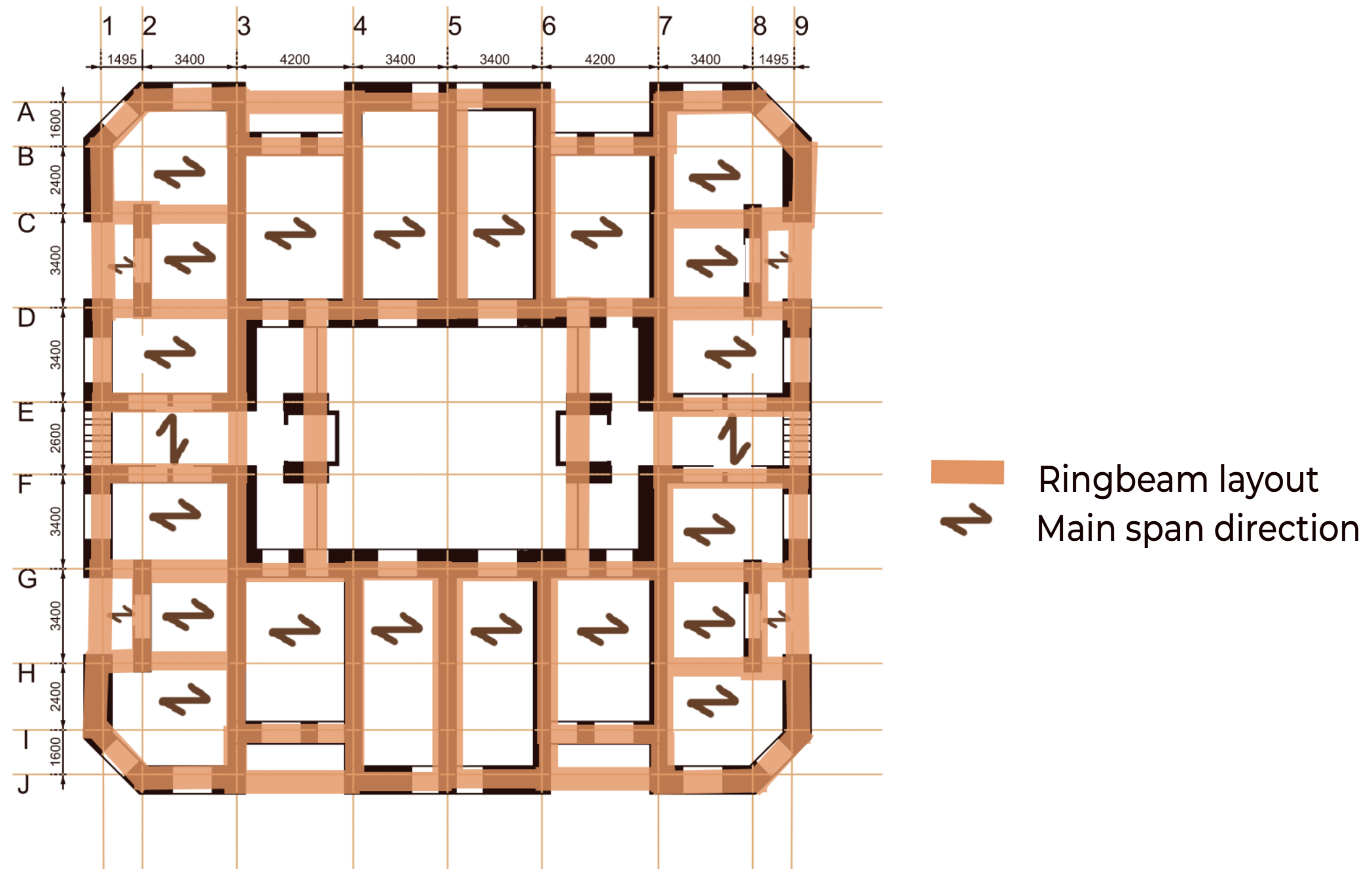
Impressions

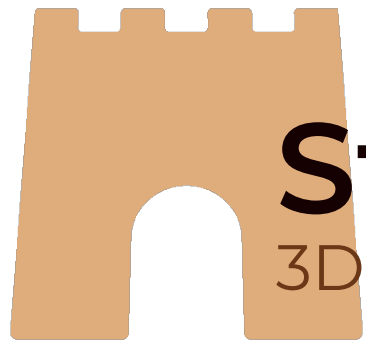
Take your moment



Structural design

Plan

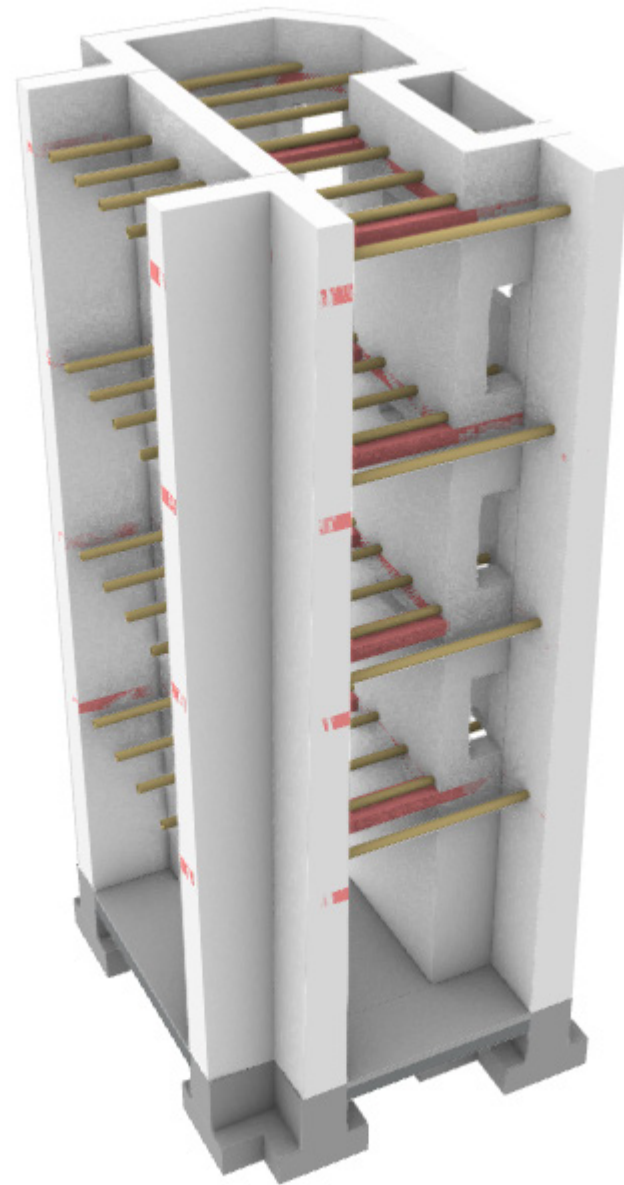




Structural design

3D impression

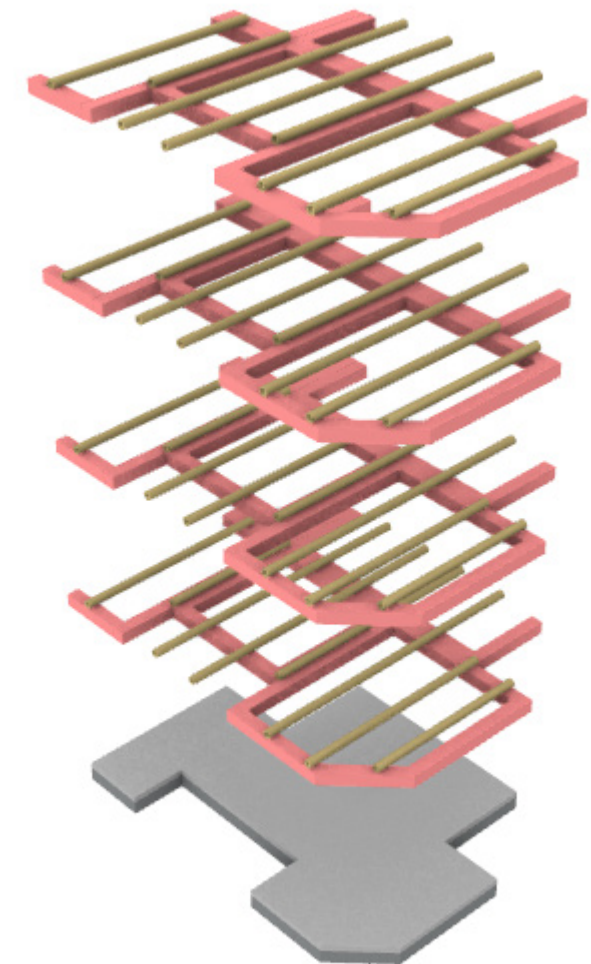
One corner of the building
Back

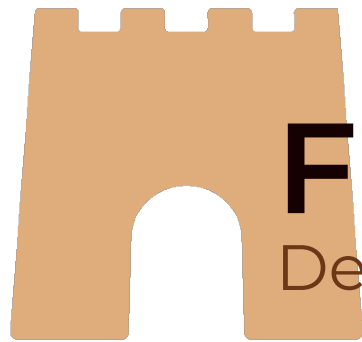


Front



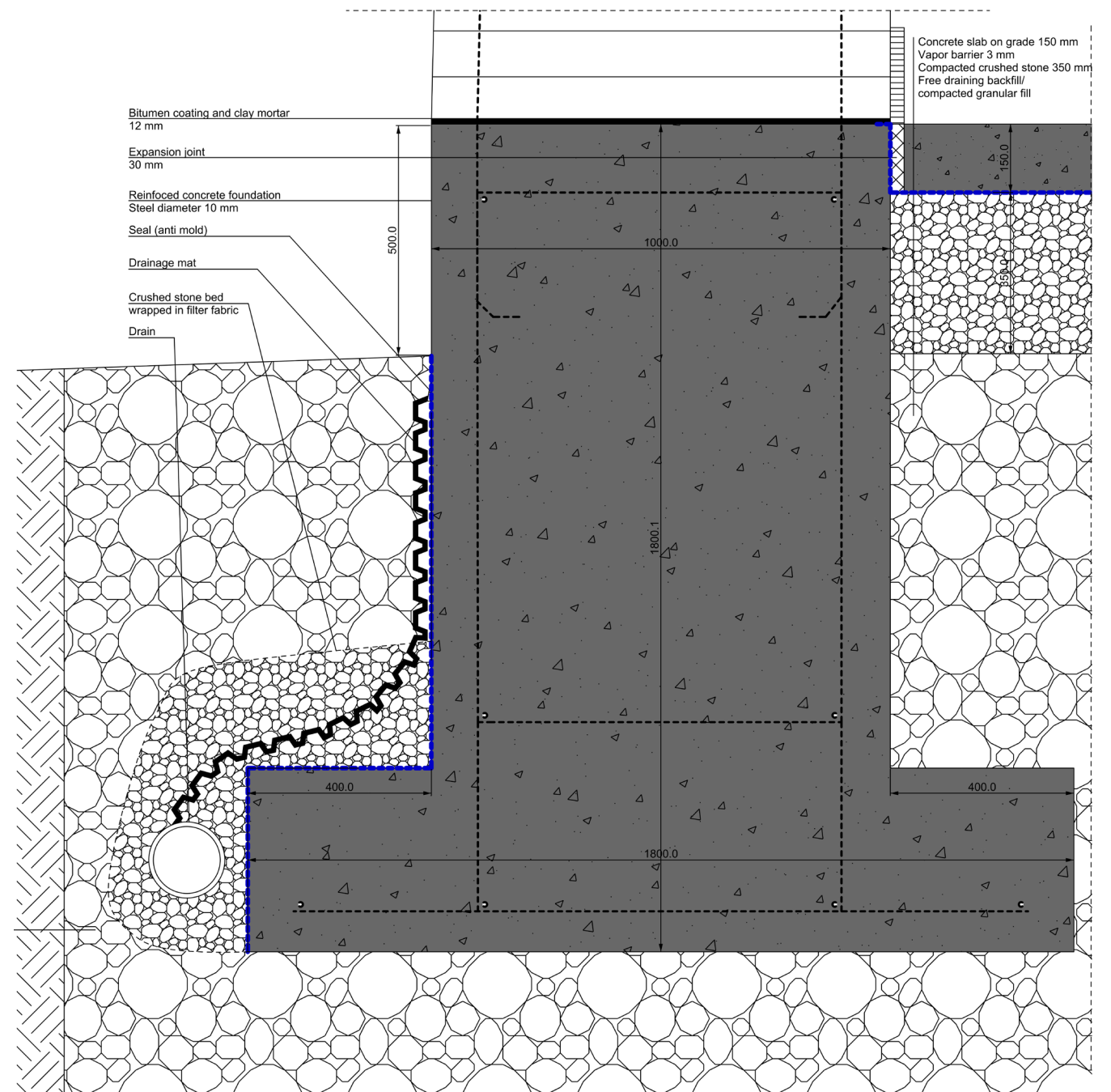
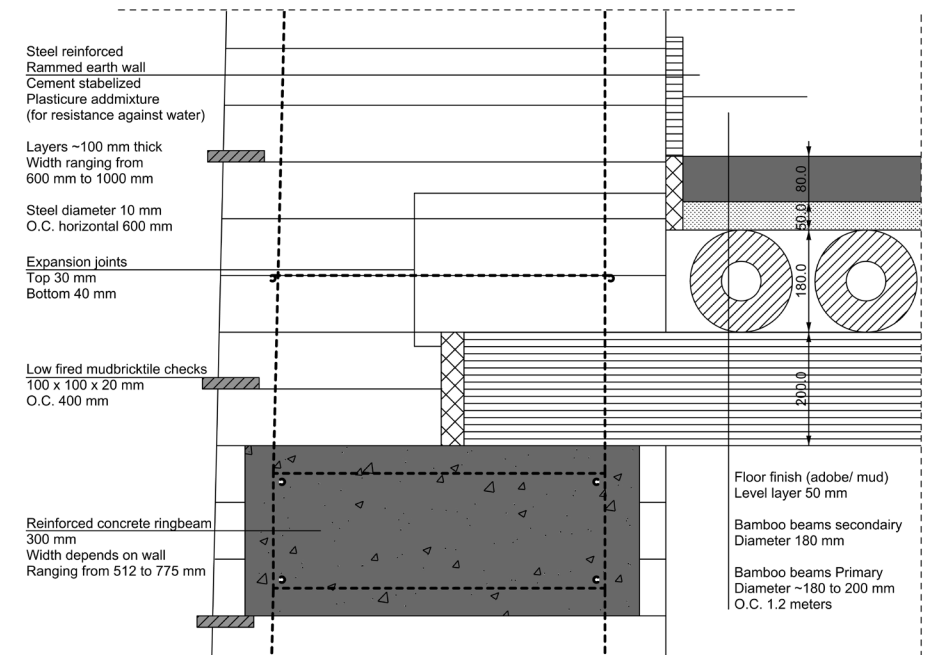
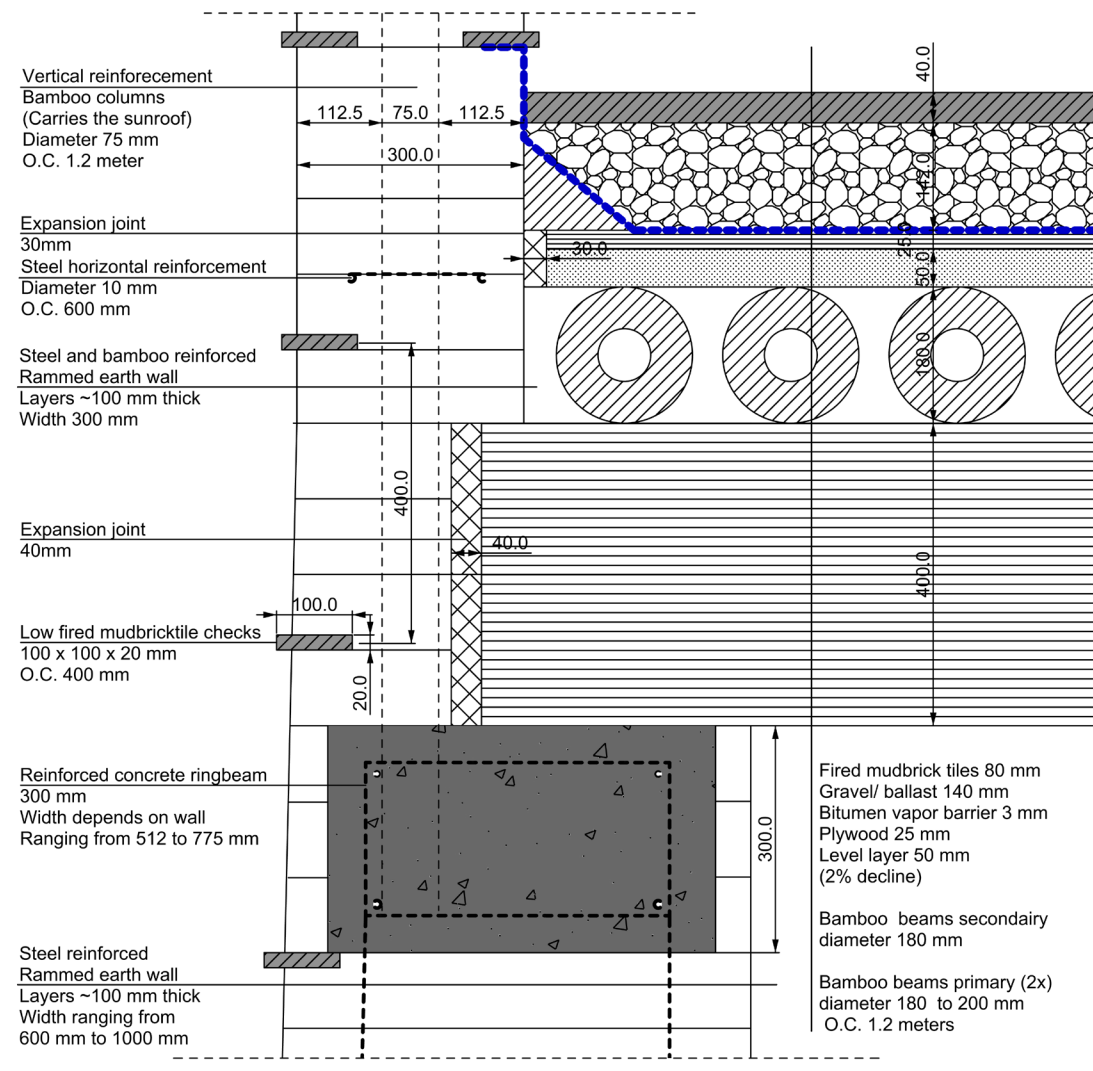
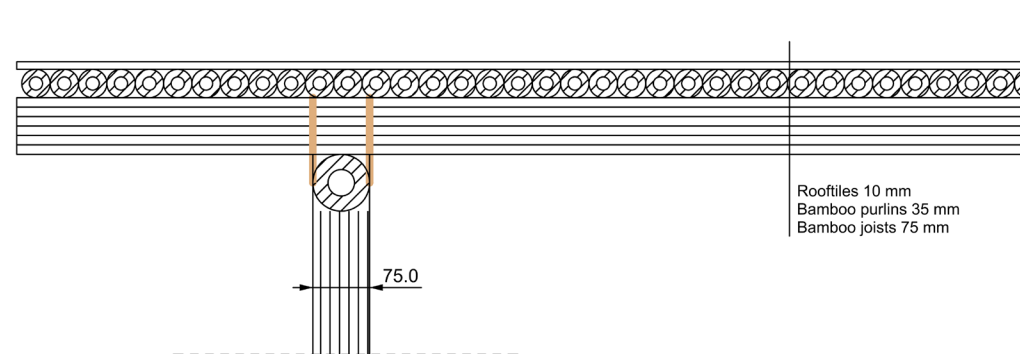
Floors only





Further detailing

Details wall element

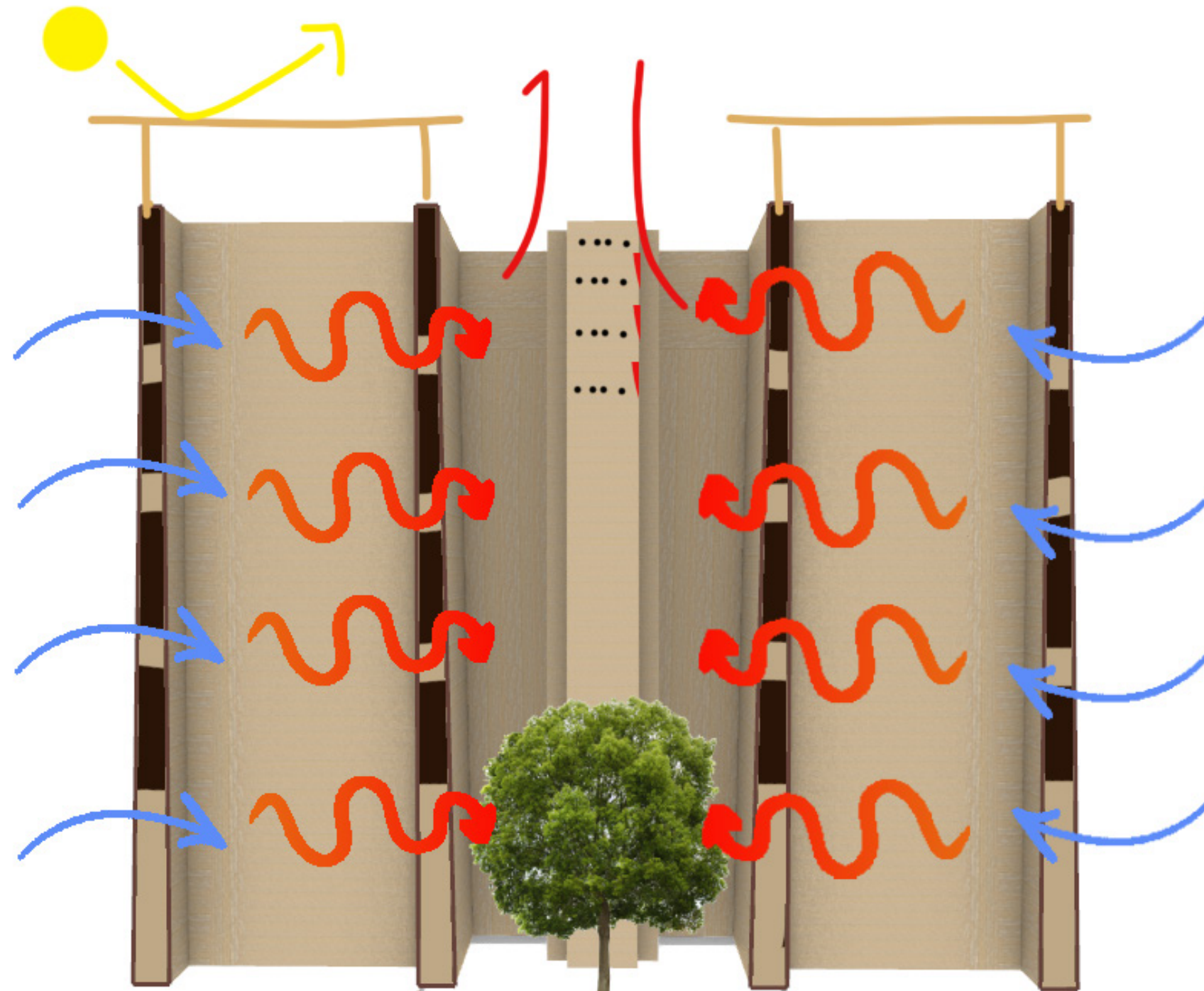




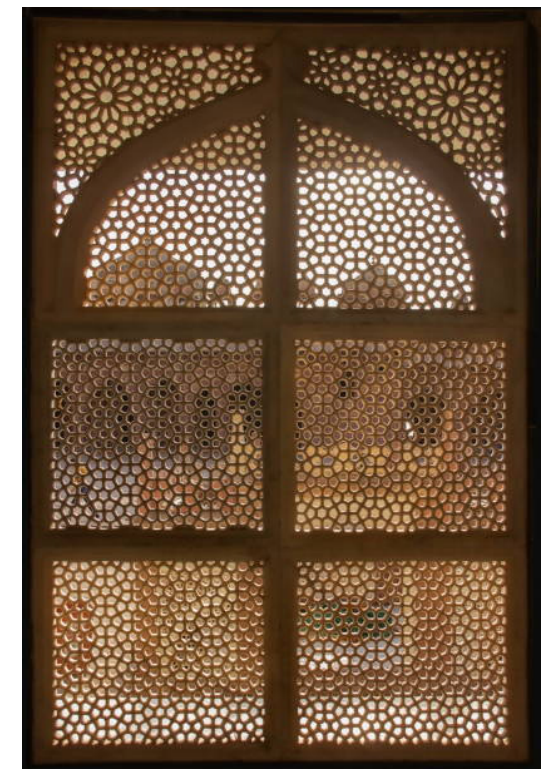
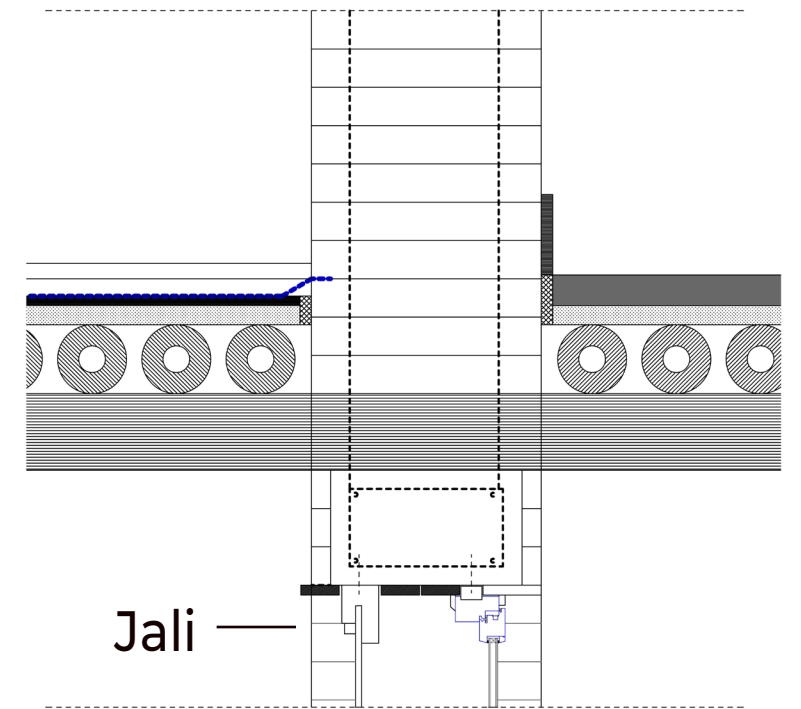
Climate design

Cilmate scheme

Section



Window detail



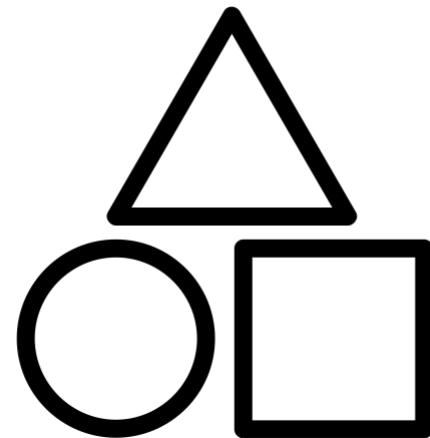


Reflection

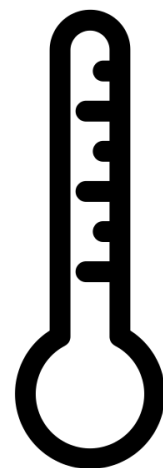
Opportunities and problems



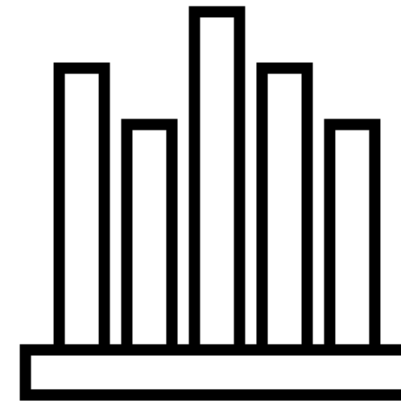
Local and sustainable



Simple



Thermal properties



Variable strength

Checks beforehand



Water and
Maintenance

Plasticure
and lime checks

